

# Implementing Wazuh at the IT Data Center of Excellence

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# Overview of Wazuh

- Open-source security monitoring platform
- Key functionalities
  - Threat Detection
  - Integrity Monitoring
  - Incident Response
  - Compliance Management
- Objectives
  - Enhance Security Posture
  - Improve Threat Detection
  - Ensure Compliance



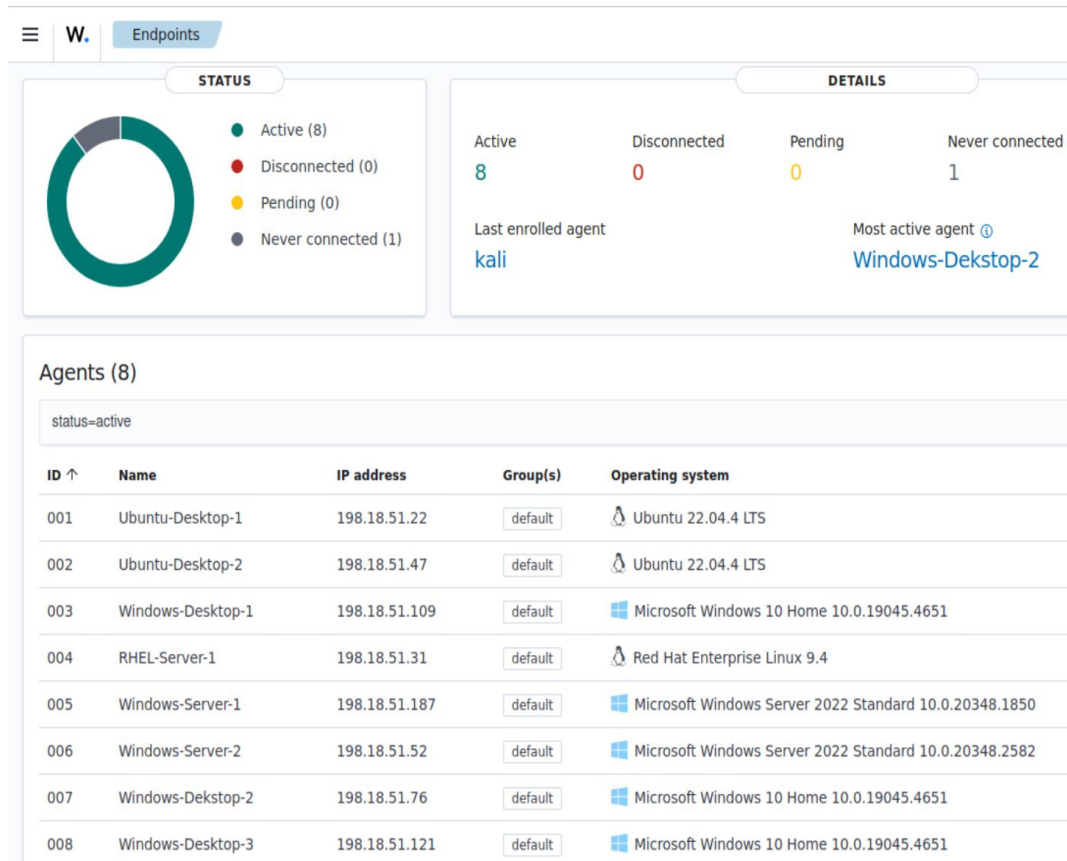


# Original Plan

- Emulate IT Center of Excellence infrastructure on a smaller scale and deploy Wazuh on all endpoints
  - 20+ clients
- Swift Deployment of Wazuh Indexer
  - Enable high scalability
- Thorough testing of a number of Wazuh capabilities
  - Wide range of VMs to be tested
- Integrate with VirusTotal

# Revised Plan

- More manageable deployment size
  - 9 clients
- Focused testing on smaller number of Wazuh capabilities and endpoints
- Several timeline adjustments





# Issues throughout the Project

- IP address for Wazuh Server/Indexer VM was not static
- VM misconfigurations hindered initial Wazuh Server setup
- Availability of macOS
- Unable to create Windows VMs for a limited time
- Some malware detection elements were inconsistent
- Integration w/VirusTotal was partially successful

```
wazuh@wazuh-virtual-machine:~$ sudo bash wazuh-install.sh --generate-config-file
s
[sudo] password for wazuh:
06/07/2024 20:22:52 INFO: Starting Wazuh installation assistant. Wazuh version:
4.8.0
06/07/2024 20:22:52 INFO: Verbose logging redirected to /var/log/wazuh-install.l
og
06/07/2024 20:22:52 INFO: Verifying that your system meets the recommended minim
um hardware requirements.
06/07/2024 20:22:52 INFO: --- Dependencies ---
06/07/2024 20:22:52 INFO: Installing systemd.
06/07/2024 20:22:53 ERROR: Cannot install dependency: systemd.
```



# Wazuh Detection and Redhat Servers

- Wazuh is an open-source security monitoring and analytics platform that provides comprehensive security monitoring capabilities.
- It helps organizations detect intrusions, monitor file integrity, analyze log data, detect vulnerabilities, and ensure compliance with security policies



# Wazuh Detection and Redhat Servers

## Benefits for Red Hat Server:

- **Enhanced Security Monitoring:** Wazuh provides real-time monitoring and alerts for potential security threats, enhancing the overall security posture of Red Hat servers.
- **Vulnerability Detection:** It regularly scans Red Hat servers for vulnerabilities, providing actionable insights to patch and secure the system promptly.
- **Compliance and Reporting:** Wazuh helps maintain compliance with security policies and regulatory requirements by continuously monitoring and generating detailed reports on the security status of Red Hat servers.



# Wazuh Detection and Redhat Servers

- What was used to find a vulnerability.
  - Kali Linux
    - Nmap 198.18.51.31 -Pn -sV
    - Sudo hping3 -V -c 10000 -d 10000 -S -p 22 -flood 198.18.51.31





# Wazuh Detection and Redhat Servers Vulnerabilities

## Issue: User Enumeration Vulnerability in OpenSSH 8.7

- Allows attackers to verify if a username and public key combination is valid.
- This occurs because a challenge is sent only when that combination could be valid for a login session

## Solution:

- **Configuration Hardening**
  - Enforce a rate limit and lockout policy
  - Use A multi-factor authentication MFA
  - Use strong, unique usernames and public keys.
- **Monitor**
  - Analyze SSH logs for suspicious patterns (in this case Wazuh)

# TheZoo

Contains 100's of  
different malwares

<https://github.com/ytisf/theZoo>

Windows 10 Desktop 2

Enforce US Keyboard Layout View Fullscreen Send Ctrl+Alt+Delete

theZoo/malware/Binaries at master · ytisf/theZoo · GitHub

ytisf db -> 1712294860000 61226e5 · 3 months ago

| Name                            | Last commit message                    | Last commit date |
|---------------------------------|----------------------------------------|------------------|
| ..                              |                                        |                  |
| AllElectroRAT                   | Revamp of malware folder + new samples | 3 years ago      |
| AndroRat_6Dec2013               | Revamp of malware folder + new samples | 3 years ago      |
| Android.PegasusB                | Revamp of malware folder + new samples | 3 years ago      |
| Android.Skygofree               | Revamp of malware folder + new samples | 3 years ago      |
| Android.Spy_49_iBanking_Feb2014 | Revamp of malware folder + new samples | 3 years ago      |
| Android.VikingHorde             | Revamp of malware folder + new samples | 3 years ago      |
| AntiExe.A                       | Revamp of malware folder + new samples | 3 years ago      |
| Artemis                         | Revamp of malware folder + new samples | 3 years ago      |
| BAT.Drop                        | Revamp of malware folder + new samples | 3 years ago      |
| BAT.Pot.A                       | Revamp of malware folder + new samples | 3 years ago      |
| RAT Skull                       | Revamp of malware folder + new samples | 3 years ago      |

Activate Windows  
Go to Settings to activate Windows

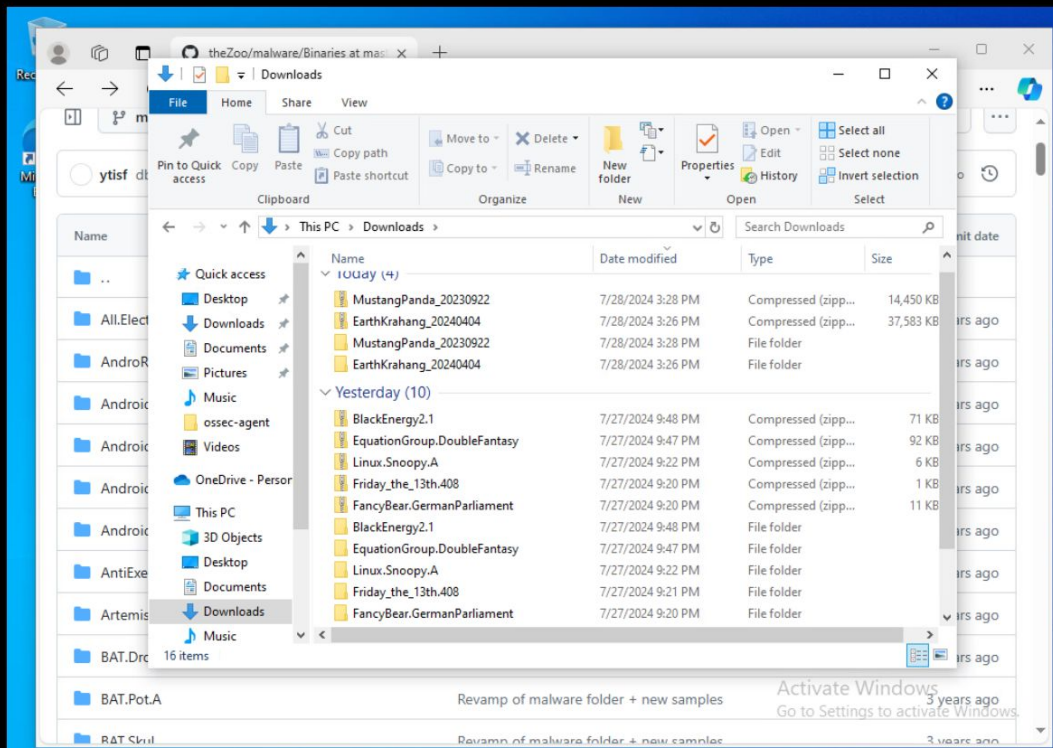
Sunday, July 28, 2024

Type here to search 96°F 3:29 PM 7/28/2024

# Malware

Windows 10 Desktop 2

Enforce US Keyboard Layout View Fullscreen Send Ctrl+Alt+Delete



Type here to search

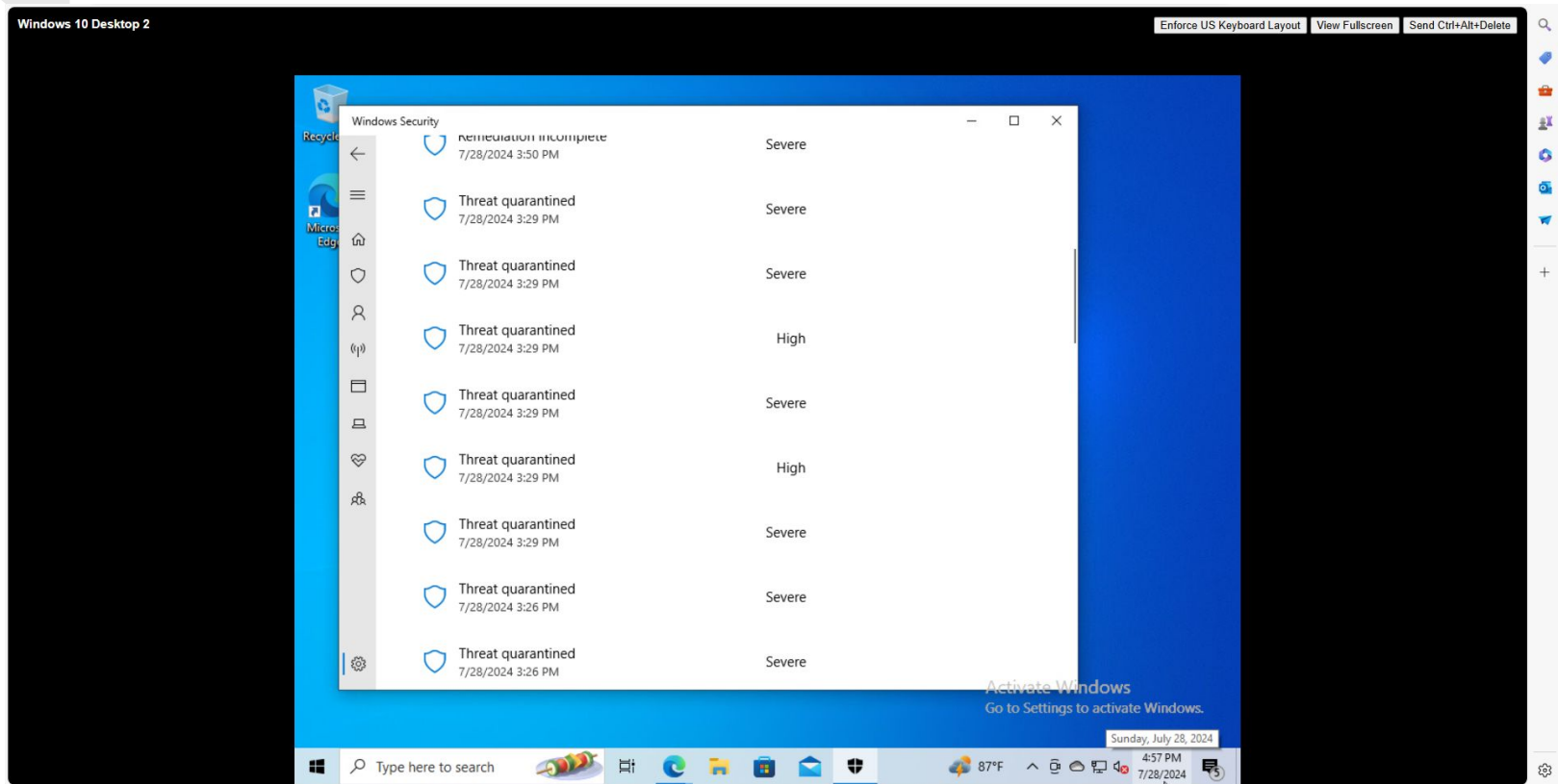


3:29 PM

7/28/2024



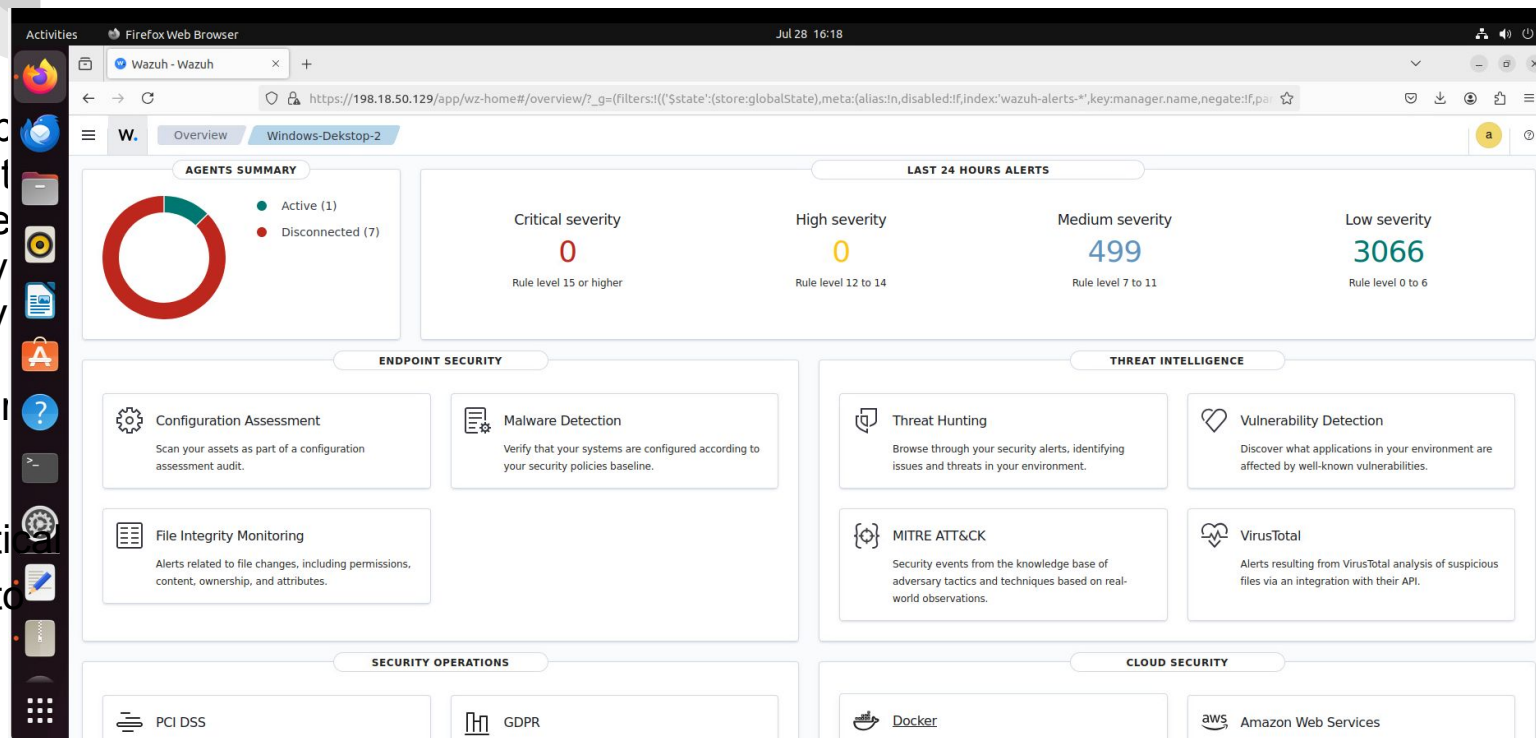
# Threats detected on Windows Server



# Wazuh Dashboard Alerts

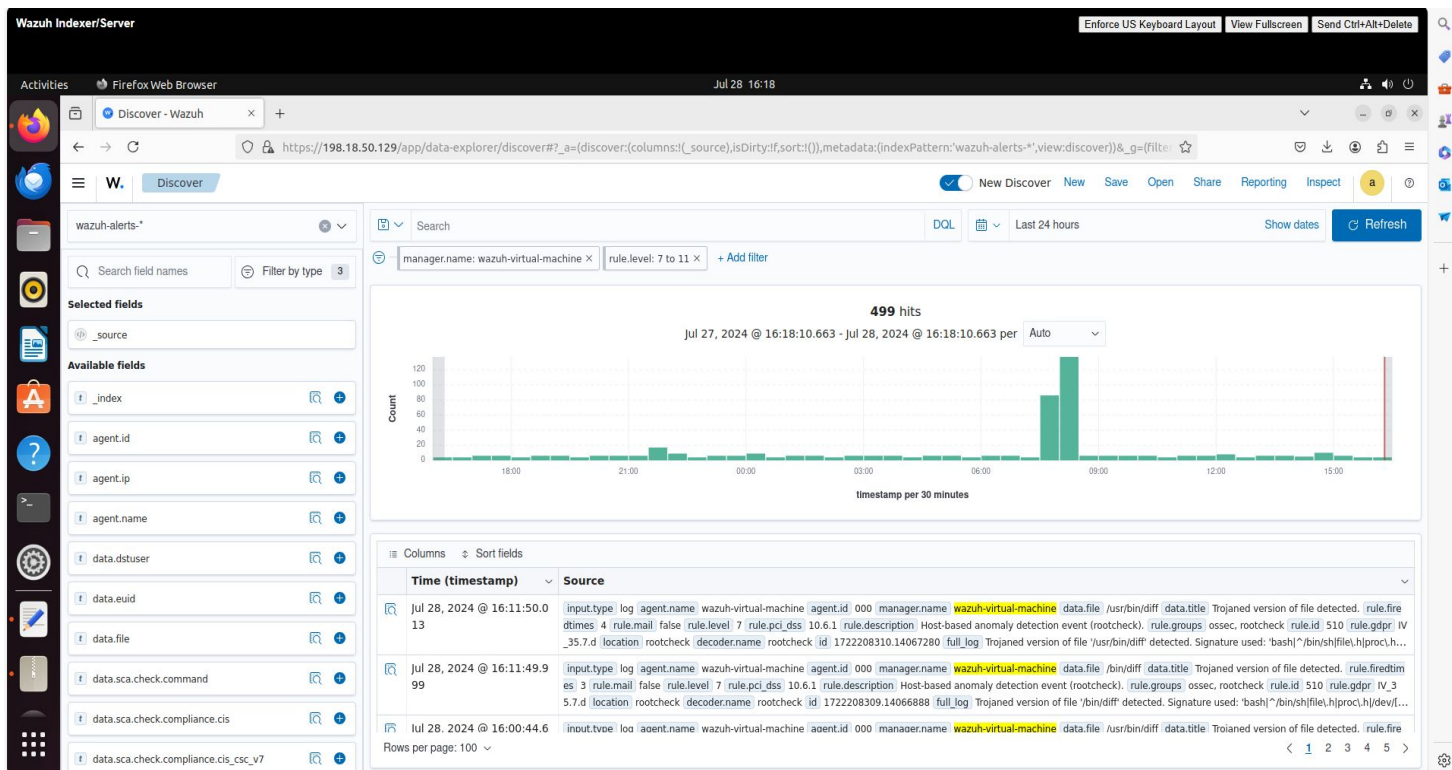
The dashboard shows all the different levels of severity of potential vulnerabilities that could show up on your servers.

From Critical Severity to Low Severity.



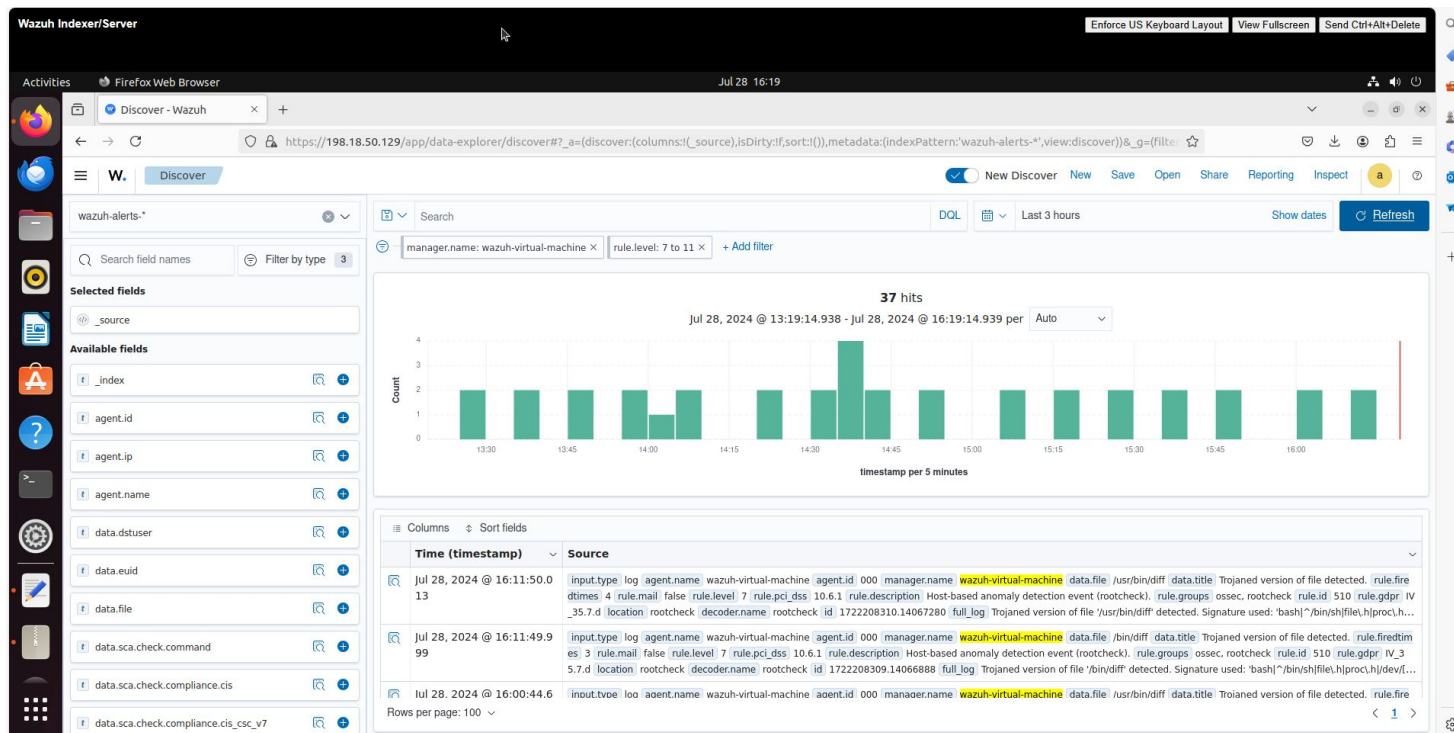
# Medium Severity Alerts (7-11)

In the past 24 hours, there have been 499 medium severity alerts from the Windows server I used.



# Viruses Detected in the Last 3 hours

There are 37  
from the  
past 2 hours



# Details of the Trojaned File that was detected

Wazuh Indexer/Server

Enforce US Keyboard Layout View Fullscreen Send Ctrl+Alt+Delete

Activities Firefox Web Browser Jul 28 16:18

Discover - Wazuh

https://198.18.50.129/app/data-explorer/discover#a=(discover:(columns!(\_source),isDirty:!f,sort!()),metadata:(indexPattern:'wazuh-alerts-\*',view:discover))&\_g=(filter

Discover

wazuh-alerts-\*

Search

manager.name: wazuh-virtual-machine x rule.level: 7 to 11 x + Add

Selected fields

Search field names Filter by type 3

Available fields

\_source

\_index

agent.id

agent.ip

agent.name

data.dstuser

data.euid

data.file

data.sca.check.command

data.sca.check.compliance.cis

data.sca.check.compliance.cis\_csc\_v7

Count

Jul 27, 2024

Columns Sort fields

Time (timestamp) Source

Jul 28, 2024 @ 16:11:50.0 13 input.type: log agent.name: wazuh-virtual-machine rule.level: 7 rule.description: Host-based anomaly detection event (rootcheck). rule.firedtimes: 4

Jul 28, 2024 @ 16:11:49.9 99 input.type: log agent.name: wazuh-virtual-machine rule.level: 7 rule.description: Host-based anomaly detection event (rootcheck). rule.firedtimes: 4

Jul 28, 2024 @ 16:00:44.6 input.type: log agent.name: wazuh-virtual-machine rule.level: 7 rule.description: Host-based anomaly detection event (rootcheck). rule.firedtimes: 4

Rows per page: 100

## Document Details

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Table JSON

|                  |                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| __index          | wazuh-alerts-4.x-2024.07.28                                                                                                           |
| agent.id         | 000                                                                                                                                   |
| agent.name       | wazuh-virtual-machine                                                                                                                 |
| data.file        | /usr/bin/diff                                                                                                                         |
| data.title       | Trojaned version of file detected.                                                                                                    |
| decoder.name     | rootcheck                                                                                                                             |
| full_log         | Trojaned version of file '/usr/bin/diff' detected. Signature used: 'bash ^/bin/sh file\\.h proc\\.h dev [^n] ^/bin/.\\*sh' (Generic). |
| id               | 1722208310.14067280                                                                                                                   |
| input.type       | log                                                                                                                                   |
| location         | rootcheck                                                                                                                             |
| manager.name     | wazuh-virtual-machine                                                                                                                 |
| rule.description | Host-based anomaly detection event (rootcheck).                                                                                       |
| rule.firedtimes  | 4                                                                                                                                     |